

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Shafiq-us Saleheen

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
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- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
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Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
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Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

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Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

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Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

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46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
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56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

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Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The recommendations consistently align with the core modules responsible for the observed faults. In each case, the suggested files point to the correct functional areas where output handling, flag validation, JSON serialization, or argument passing occur.

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Even when more than one method could be involved, the suggested methods remain within the correct execution path and direct attention to the code that controls output generation, error reporting, and configuration handling. In practice, a developer starting from INSIGHT's method suggestions would be guided to the correct call sites and logic branches.

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

When it proposes buggy files and methods, it justifies them using details such as where output is routed, where flags and meanings are validated. Even when its localization is somewhat broad, the rationale remains consistent with the architecture of the project.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

It is not always clear that these suggestions are tightly grounded in the specific design choices, invariants, and testing patterns of this particular repository. In several cases, the recommendations read more like generic best practices.

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution ^{*} features were?

- ☒ Very accurate
- ☐ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

How helpful is INSIGHT in terms of localizing buggy code files and methods? ^{*}

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☒ Very responsive
- ☐ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

Call-graph analysis, cross-file dependency tracing, and repository-specific fix recommendations that automatically map detected issues to precise code regions and relevant tests.

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

6

How many years of experience do you have in using Github?

6

How many years of experience do you have in managing and solving issue reports?

3

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Google Forms

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<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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An apparently failed result, with no reasoning message.

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Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```


Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The files which should contain bugs are in the comments given by insight for most issues

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

It found the file , but sometimes giving different method in the same file. But some methods were correctly identified

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Relevant with the found bugs

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Steps were clear

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

Good

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

8

How many years of experience do you have in using Github?

8

How many years of experience do you have in managing and solving issue reports?

6

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Abu Rifat Muhammed Al Hasib

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on '**New issue**'
- (d) Copy and paste the content in the title and description sections
- (e) Click '**Submit new issue**'

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns          Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It found the files

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Most functions seems to related to the bug

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It analyzed the results properly

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Suggests good approach

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Show the code snippets would be better too

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
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Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☒ Very responsive
- ☐ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
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What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

Great set of features

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

5

How many years of experience do you have in using Github?

5

How many years of experience do you have in managing and solving issue reports?

2

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

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If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

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Please Enter your Name *

Muhammad Shazzad Mia

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Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

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Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

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```
46: """
47:     Static check runner.
48:
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50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:Reasoning
------	-----------	-----------------

§3.5 Flags and flag attributes	:3:	31/32 :
--------------------------------	-----	---------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```


Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
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- (c) Click on **'New issue'**
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Found the files which contained bugs

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Found most methods

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Well explained

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Helpful to debug the files

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
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Please specify the reason for your answer (optional).

.....

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What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

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Add fix generation

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Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

it's good

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

it's good

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

it's good

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☐ Moderately good
- ☒ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

it's good

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☐ Moderately accurate
- ☒ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☐ Moderately helpful
- ☒ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☐ Moderately helpful
- ☒ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☐ Moderately responsive
- ☒ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

nothing for now

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

3

How many years of experience do you have in using Github?

5

How many years of experience do you have in managing and solving issue reports?

3

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Mustahid Hasan

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```


Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The tool can partially identify issue 1.

GT: ['compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Generated: ComplianceChecker in compliance_checker/runner.py

check_errors in compliance_checker/runner.py

_print_checker in compliance_checker/suite.py

Issue 02:

The tool covers all the GT buggy files and additionally adds "_check_flag_values" which is not in GT. Technical analysis seems correct.

Issue 03:

GT: ['compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Generated: html_output in compliance_checker/suite.py

_get_check_versioned_name in compliance_checker/suite.py

html_output in compliance_checker/runner.py

Comments: Seems like Insight's analysis on localizing ground truth seems correct rather than ground truth.

Issue 04:

GT files: ['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Insight: html_output in compliance_checker/suite.py

run_all in compliance_checker/suite.py

run_checker in compliance_checker/runner.py

Comments: Insight misses test_cli but correctly recognizes suite.py files. And analyses are based on their localized files. Seems like 50% similarity according to ground truth.

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Somewhat. In some cases, the methods in the ground truth and generated files seem incorrect. But, in most cases, it correctly recognizes the buggy files.

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It clearly explains the step-by-step analysis, just like the ground truth data. However, where they miss the important files at first, they miss the explanation for this.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It clearly explains the step-by-step analysis,

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

NA

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution * features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

Answered

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☒ Very responsive
- ☐ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

After rigorous testing against a large dataset, the tool's automatic patch suggestions will be handy.

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☐ Bachelors
- ☐ Masters
- ☒ PhD
- ☐ Others

How many years of experience do you have in programming?

8

How many years of experience do you have in using Github?

8

How many years of experience do you have in managing and solving issue reports?

5

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

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The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

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CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

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Mouneeta Rahman

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Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
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- (d) Copy and paste the content in the title and description sections
- (e) Click '**Submit new issue**'

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Most of the files were relevant or close enough

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

Most methods seemed to be missing from the results

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
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Please explain your answer (why yes, no, or unsure?)

Explains properly

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☒ Very easy
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How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Since I could not iterate through the approaches, cannot verify. But the structure seems to be fine.

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
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How easy or difficult to use INSIGHT is? *

- ☒ Very easy
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Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
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Please explain your answer (optional).

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How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
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- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

It would have been better if the app was interactive. If it could try to localize again , according to my instructions

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
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- ☐ Masters
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Md Arif Hasan

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Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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...

An apparently failed result, with no reasoning message.

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Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

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Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT successfully identified most of the key files involved in the issues, and its suggestions were generally aligned with the actual buggy locations.

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT highlighted relevant methods and provided logical reasoning that matched the structure of the codebase.

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT's analysis was generally helpful and showed good understanding of the repository, but in a few places it was not fully clear whether all contextual details were captured.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☒ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☐ Moderately good
- ☒ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

None

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Most suggestions aligned well with the actual buggy areas, though a few cases felt slightly generic.

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution * features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

Most suggestions aligned well with the actual buggy areas, though a few cases felt slightly generic.

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

It pointed to the correct regions in many cases, but some suggestions could be more precise.

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☐ Moderately helpful
- ☒ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

The repair strategies followed common debugging practices but sometimes lacked deeper code-specific detail.

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☐ Moderately responsive
- ☒ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☒ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

Provide explanations with more concrete code examples and integrate side-by-side diff views for easier understanding.
.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☐ Bachelors
- ☒ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

5

How many years of experience do you have in using Github?

5

How many years of experience do you have in managing and solving issue reports?

2

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Lutfar Rahman Alif

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

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Recommended Fix:

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File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

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```
46: """
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49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
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53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format Format of the output(s) (e.g., 'text', 'html', 'json').
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58: @returns Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
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Fail result for Flags and flag attributes check with no reasoning

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Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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[check_flags]

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Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

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1213:         if not allvr.value:
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1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
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JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

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compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

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[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
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Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:                 cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

because the suggestions it produced pointed to the components involved in the actual root cause of the issue.

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

because the methods it highlighted aligned with the core execution flow responsible for the issue.

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT's technical analysis demonstrated that it understood the structure and flow of the repository's codebase while identifying the buggy files and methods.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☐ Moderately good
- ☒ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It accurately aligned with the standard and effective way developers resolve this type of bug.

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☒ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

N/A

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

3

How many years of experience do you have in using Github?

5

How many years of experience do you have in managing and solving issue reports?

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Arnab Das Joy

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns          Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Correctly identified the relevant files for each issue. Its suggestions concurs with the expected buggy files like runner.py, test_cli.py, cf.py, cf_1_6.py, and util.py

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The tools correctly identified the core methods potentially responsible for the bugs. Methods like `run_checker`, `json_output`, `check_flags`, and `download_cf_standard_name_table`. These match the expected buggy methods referenced e.g., `ComplianceChecker`, `TestCLI`, `check_flags`.

.....

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The analysis provided an understanding of how different modules interact. Specifically output handling in runner.py, flag validation logic in cf.py, and JSON formatting components.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The repair suggestions aligned well with typical resolutions like redirecting stdout properly, adding missing parameters to method calls, suppressing debug output in JSON mode, and improving error messages in flag checks. These match the expected solutions provided.

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
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Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution ^{*} features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

How helpful is INSIGHT in terms of localizing buggy code files and methods? ^{*}

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
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What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

N/A

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

3.5

How many years of experience do you have in using Github?

3

How many years of experience do you have in managing and solving issue reports?

0.5

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

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CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

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Please Enter your Name *

Jagannath Biswas

Overview of INSIGHT

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- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
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Save text output

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I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

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Buggy Files:

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`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
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54: @param skip_checks    Names of checks to skip
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58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
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Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:Reasoning
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§3.5 Flags and flag attributes	:3:	31/32 :
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An apparently failed result, with no reasoning message.

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Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
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JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

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Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

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Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It seemed to properly find the files where bugs are located

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

Maybe not all functions are covered

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Technical analysis matches the files and functions

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The fixes align with the bug in the lines , and gives a result. It should give developers an idea on how to fix it

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☒ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

More features needed to make issue resolving easy

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☐ Bachelors
- ☒ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

3

How many years of experience do you have in using Github?

4

How many years of experience do you have in managing and solving issue reports?

2

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

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<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Identified very uncommon bug

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Not sure how it did

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☒ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

Not familiar with these

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☐ Moderately good
- ☒ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

Not familiar

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☐ Moderately accurate
- ☒ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☐ Moderately helpful
- ☒ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

Best of luck
.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

0

How many years of experience do you have in using Github?

1

How many years of experience do you have in managing and solving issue reports?

0

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Md. Saleh Rafi

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on '**New issue**'
- (d) Copy and paste the content in the title and description sections
- (e) Click '**Submit new issue**'

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT correctly identified the relevant methods (e.g. run_checker, check_flags, json_output). These were the original source of the bug, which is consistent with Ground Truth.

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT analyzed the context of the codebase well. It was able to accurately capture issues such as CLI behavior, JSON output handling, error messaging, and method signature mismatch.

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT accurately analyzed the codebase context of the repository. It was able to identify the correct files and methods by understanding CLI workflow, JSON output handling, error propagation, and method call dependencies. INSIGHT's technical analysis clearly explained how bugs propagated across functions, which was consistent with Ground Truth.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The Recommended Fixes that INSIGHT provides are accurate and practical. The suggested solutions directly target the root cause of the bug—such as adding missing arguments, separating debug information in JSON output, and improving documentation and error messages. These suggestions are consistent with Ground Truth and are effective in solving bugs in practice.

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

No

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution ^{*} features were?

- ☒ Very accurate
- ☐ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

How helpful is INSIGHT in terms of localizing buggy code files and methods? ^{*}

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☒ Very responsive
- ☐ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☒ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)
- ☒ Other: Showing a live code snippet preview with Recommended Fix will increase usability.

Please provide any (other) recommendations to improve INSIGHT

No

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

3 years

How many years of experience do you have in using Github?

3 years

How many years of experience do you have in managing and solving issue reports?

1 year

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

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CONSENT

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Please Enter your Name *

Sheikh Sarafat Hossain

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Save text output

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I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

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Buggy Methods:

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Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
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50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
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53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
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56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns          Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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...

An apparently failed result, with no reasoning message.

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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
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- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
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- (c) Click on **'New issue'**
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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT was able to consistently identify the correct files related to the reported issues.

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT identified the correct methods or classes responsible for the bugs.

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It correctly identified the roles of key methods such as stdout_output and run_checker, explaining how output is currently managed and why file redirection is limited to CLI-based mechanisms.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT's recommended fix accurately addresses the root cause of the reported issue.

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☒ CI/CD integration (post findings on PRs, block risky merges, regression checks)
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- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

Impact analysis (how a reported bug affects other components, dependent modules, or end users)
.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
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```
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Issue Description:

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<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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...

An apparently failed result, with no reasoning message.

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- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:                 cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

For most of the issues it did a good job at localizing bug but for only one issue it failed to locate i think.

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

For method level, it wasnt very precise and didnt detect the exact bug but it did a very good analysis job for the possible methods that can have bug with justification and i think it was good enough.

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

As i have less context about the system but after reading the analysis it seemed easy to understand the methods and have a good understanding to localize the bug from the suggested methods.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

As it suggests recommended fix for all the possible methods that can have bug, for that i need to examine all the suggested methods and code chunk to localize the actual bug because INSIGHT is not that precise at localizing the actual bug.

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution * features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☒ Very responsive
- ☐ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

I am not sure if i can further communicate with the bot to locate the actual bug from the suggested methods but that could be helpful and time saving.

Participant's Background and Experience

Please choose your highest credit obtained

- ☒ High School
- ☐ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

3

How many years of experience do you have in using Github?

2

How many years of experience do you have in managing and solving issue reports?

0

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Rijia Parveen Raya

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:Reasoning
------	-----------	-----------------

§3.5 Flags and flag attributes	:3:	31/32 :
--------------------------------	-----	---------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

- ☐ Yes
- ☒ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

INSIGHT did not provide clear or observable file-level bug localization suggestions after the issue was created.

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

did not provide any observable method-level bug localization suggestions after the issues were created

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It did not provide a clear or observable technical analysis explaining how the buggy files and methods were localized within the context of the repository.

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☐ Moderately good
- ☒ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

unsure

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

The interface is easy to use. However, it would be helpful to provide clearer visual feedback after issue submission, such as analysis status indicators or notifications when results are ready.

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution ^{*} features were?

- ☐ Very accurate
- ☐ Moderately accurate
- ☒ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

How helpful is INSIGHT in terms of localizing buggy code files and methods? ^{*}

- ☐ Very helpful
- ☐ Moderately helpful
- ☒ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☐ Moderately responsive
- ☐ Neutral
- ☒ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☒ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)

☐ Other:

Please provide any (other) recommendations to improve INSIGHT

while INSIGHT is easy to use, improving result visibility, reliability, and feedback mechanisms after issue submission would significantly enhance its practical usefulness.

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

1–2 years

How many years of experience do you have in using Github?

1 year

How many years of experience do you have in managing and solving issue reports?

0 years

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

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Md Al Amran

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Save text output

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I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

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Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns          Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
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- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It gives details of the positions where bug localization might be happened

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

maybe not all methods are covered

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Explanation are given properly

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Suggested fix are there, but short

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
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How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
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Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
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Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
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.....

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- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

More details in patch suggestion expected

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- ☐ High School
- ☒ Bachelors
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Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:                 cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It found the files where bugs are

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

It found some files and missed some i guess

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The analysis is well written

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

The fixes are relevant to the technical analysis

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☒ Very responsive
- ☐ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

The patch generation feature should be improved

Participant's Background and Experience

Please choose your highest credit obtained

- ☒ High School
- ☐ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

4

How many years of experience do you have in using Github?

4

How many years of experience do you have in managing and solving issue reports?

2

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Mashrief Bin Zulfiquer

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on '**New issue**'
- (d) Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns          Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

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- Go to the repository
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Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

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Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Yes

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

Unsure

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

Yes

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

Unsure

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☒ Very accurate
- ☐ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
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Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
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What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
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Please provide any (other) recommendations to improve INSIGHT

All good

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
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- ☒ Masters
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How many years of experience do you have in programming?

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Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

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Ankan Shahriar Islam

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Save text output

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I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

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Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:                 cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It explains properly

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

It localizes the files

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
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Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

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- ☐ No
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Please explain your answer (why yes, no, or unsure?)

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- ☐ Very easy
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- ☒ Moderately good
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Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
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- ☐ Very accurate
- ☒ Moderately accurate
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I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

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After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on '**New issue**'
- (d) Copy and paste the content in the title and description sections
- (e) Click '**Submit new issue**'

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

.....

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☒ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☐ Bachelors
- ☒ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

6

How many years of experience do you have in using Github?

5

How many years of experience do you have in managing and solving issue reports?

3

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Mushfiqur Rahman

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on '**New issue**'
- (d) Copy and paste the content in the title and description sections
- (e) Click '**Submit new issue**'

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:Reasoning
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§3.5 Flags and flag attributes	:3:	31/32 :
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...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

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How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
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Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

.....

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☐ Moderately helpful
- ☒ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

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How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
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What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

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- ☐ High School
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How many years of experience do you have in programming?

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3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:Reasoning
------	-----------	-----------------

§3.5 Flags and flag attributes	:3:	31/32 :
--------------------------------	-----	---------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

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Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
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Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

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Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
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Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

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Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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...

An apparently failed result, with no reasoning message.

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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
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- (c) Click on **'New issue'**
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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☐ Moderately easy
- ☒ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☒ Very accurate
- ☐ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

4

How many years of experience do you have in using Github?

3

How many years of experience do you have in managing and solving issue reports?

3

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Sadia Tabassum

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
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Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

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46: """
47:     Static check runner.
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list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
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```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format Format of the output(s) (e.g., 'text', 'html', 'json').
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Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
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<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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...

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- Go to the repository
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Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
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1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
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Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

.....

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☒ Very accurate
- ☐ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
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- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☒ Very responsive
- ☐ Moderately responsive
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What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
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.....

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Please choose your highest credit obtained

- ☐ High School
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- ☒ Masters
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How many years of experience do you have in programming?

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Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

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Nazmus Sakib

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In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
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- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on '**New issue**'
- (d) Copy and paste the content in the title and description sections
- (e) Click '**Submit new issue**'

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:Reasoning
------	-----------	-----------------

§3.5 Flags and flag attributes	:3:	31/32 :
--------------------------------	-----	---------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
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Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

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- ☐ Neutral
- ☐ Moderately poor
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Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

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features were?

- ☒ Very accurate
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Nurul Islam

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- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
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- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

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- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
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Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
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- There are no right or wrong answers—we value your honest feedback
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Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

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Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns          Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:Reasoning
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§3.5 Flags and flag attributes	:3:	31/32 :
--------------------------------	-----	---------

...

An apparently failed result, with no reasoning message.

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1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

.....

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☐ Moderately accurate
- ☒ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

4

How many years of experience do you have in using Github?

4

How many years of experience do you have in managing and solving issue reports?

3

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Md Arafat Hossain

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

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- Create a new issue in the test repository
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(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
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- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
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- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
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Buggy Files:

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Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
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50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
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```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
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58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

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Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:	Reasoning
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§3.5 Flags and flag attributes	:3:	31/32 :	...
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- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
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Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

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Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
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Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☒ Very accurate
- ☐ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☐ Moderately responsive
- ☒ Neutral
- ☐ Moderately unresponsive
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What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☒ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
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- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

4

How many years of experience do you have in using Github?

3

How many years of experience do you have in managing and solving issue reports?

1

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Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

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Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Noushin Tabassum

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on '**New issue**'
- (d) Copy and paste the content in the title and description sections
- (e) Click '**Submit new issue**'

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
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- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
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2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level bug localization for the created issues? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
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Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

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How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☒ Very easy
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How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
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Evaluating the Overall Experience with INSIGHT

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Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

.....

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☒ Very accurate
- ☐ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
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Please explain your answer (optional).

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How helpful is INSIGHT in terms of localizing buggy code files and methods? *

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Nazmus Salehin

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- **Copy and paste** the provided **issue title** and **description** exactly as given
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Evaluation of INSIGHT

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I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

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Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

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53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
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56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
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Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

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3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

.....

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☐ Moderately helpful
- ☒ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
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- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☐ Bachelors
- ☒ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

5

How many years of experience do you have in using Github?

4

How many years of experience do you have in managing and solving issue reports?

1

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Google Forms

Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Afnan Jarif

Overview of INSIGHT

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Function: run_checker

Lines: 46-59

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<i>Name</i>	<i>Priority:</i>	<i>Score:Reasoning</i>
-------------	------------------	------------------------

<i>§3.5 Flags and flag attributes</i>	<i>:3:</i>	<i>31/32 :</i>
---------------------------------------	------------	----------------

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[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

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Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

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```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

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IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

How easy is it to understand INSIGHT's method-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☐ Very good
- ☒ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

.....

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☒ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☐ Very responsive
- ☒ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☐ Automatic PR generation with fixed patch
- ☐ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☒ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☒ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☐ Bachelors
- ☒ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

6

How many years of experience do you have in using Github?

4

How many years of experience do you have in managing and solving issue reports?

2

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Evaluating INSIGHT, a GitHub Plugin for Bug Localization and Repairing Suggestions

RESEARCH GOAL AND PROCEDURE

The goal of this study is to evaluate INSIGHT, an automated repository-aware bug localization assistant for GitHub. INSIGHT uses a multi-stage RAG pipeline, combining dense retrieval, graph-based enrichment, and LLM reasoning to suggest buggy files and functions for a reported issue. Moreover, INSIGHT provides LLM-inferred general ways to address such bugs after localization.

PARTICIPATION AND CONFIDENTIALITY

You must be at least 18 years old to participate.

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission. Your responses to this study will be kept confidential. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty.

CONTACT

If you have any questions, please ask us at any moment. If you have any additional questions later, Jitesh Sureka (bsse1115@iit.du.ac.bd, +8801780935761), Ahmed Adan (bsse1131@iit.du.ac.bd, +8801813865290), Proma Chowdhury (bsse1132@iit.du.ac.bd, +8801767895677) and Dr. Kazi Muheymin-us Sakib (sakib@iit.du.ac.bd, +8801730051232), will be happy to answer them.

CONSENT

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE.

IF YOU WANT TO PARTICIPATE, PLEASE ENTER YOUR NAME IN THE TEXT FIELD BELOW, AND START THE SURVEY.

Please Enter your Name *

Shahriar Hossain Samir

Overview of INSIGHT

INSIGHT is an intelligent GitHub assistant that automatically analyzes bug reports and suggest code functions that likely contain the issue. When developers report a bug, INSIGHT understands the relationships between different parts of the code, and provides specific suggestions about what needs to be fixed, potentially saving hours of manual debugging time.

Study Overview

You are provided with a test GitHub repository, with the **INSIGHT** tool already installed in it, and a set of issue reports describing real-world software bugs for that repository. For each issue report, you will create a new issue in the repository, and evaluate INSIGHT's suggestions, including bug localization in the relevant files and functions, technical analysis of why the bug is present within a particular file(s)/method(s), and recommended fix for the bug. You will then compare INSIGHT's suggestions against the developer-confirmed Ground Truths assess INSIGHT's predictive accuracy, usability, usefulness, and contextual capability.

Based on your experience with INSIGHT, you will answer a set of questions to give us feedback on how well the tool performs in localizing bugs and provide suggestions to improve it.

In summary, you will need to perform the following steps (detailed instructions are provided later in this study):

1. Access the Test Repository:

- **INSIGHT** is already installed in the test repository that we are providing
- Visit the repository: [\[Test Repository Link\]](#)
- Familiarize yourself with the repository structure

2. For each of the provided issues in this survey, you will need to perform the following:

(a) Read and Understand the Issue:

- Carefully read the provided issue report with its title and description
- Review the Ground Truth (GT) information provided here showing which files/functions actually contain the bug
- Understand the nature of the bug being reported

(b) Submit the Issue:

- Create a new issue in the test repository
- **Copy and paste** the provided **issue title** and **description** exactly as given
- This step mimics the real-world scenario when a developer discovers a problem and reports it via the issue tracker
- Wait for INSIGHT to process the issue (typically takes **10-20 seconds**)

(c) Analyze INSIGHT's Response:

- Examine INSIGHT's automated comments under the created issue.
- Take notes about INSIGHT's suggested buggy files, functions, technical analysis of the bug, and the provided suggestions regarding how practitioners commonly fix these bugs.
- Compare INSIGHT's provided suggestions with our provided Ground Truth (information about the actual buggy files, methods, bug analysis, and recommended fix).

3. Provide Overall Feedback:

- After completing all bug localization tasks, answer questions about your overall experience with INSIGHT
- Share your thoughts on the tool's strengths, weaknesses, and potential improvements
- Answer questions regarding your professional background and experience with debugging

Important Notes

- Feel free to explore INSIGHT's suggestions in detail and click on the provided links to view the actual code
- You may spend as much time as needed to thoroughly evaluate each response
- There are no right or wrong answers—we value your honest feedback
- If INSIGHT fails to provide suggestions or encounters an error, please note this in your responses

Thank you for participating in this study. Your feedback will help us improve INSIGHT and make it more useful for software developers!

Evaluation of INSIGHT

Issue Report 1:

Please read and understand the following issue report:

Issue Title:

Save text output

Issue Description:

I noticed that simple text outputs are only redirected to `stdout`. There is on API to save it to a file. Correct me if I am wrong but I could not see it [here](https://github.com/ioos/compliance-checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L43-L50).

I do know that in CLI interface it is easy to redirect to a file but it would be nice to be able to do that from class method too. Not also that the docs give the idea that this is possible and the error message [here](https://github.com/ioos/compliance-

checker/blob/7276abba102e2ef6cb390d49ba0c38726f625913/compliance_checker/runner.py#L53) is not very helpful.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

`['compliance_checker/tests/test_cli.py', 'compliance_checker/runner.py']`

Buggy Methods:

`['ComplianceChecker', 'TestCLI']`

Recommended Fix:

Summary: Improve documentation and error messages in run_checker function for better clarity on output redirection and handling of missing checks.

File: compliance_checker/runner.py

Function: run_checker

Lines: 46-59

Fix: Improve documentation for output redirection in run_checker function.

Suggested Fix:

```
46: """
47:     Static check runner.
48:
49:     @param ds_loc      Dataset location (url or file)
50:     @param checker_names List of string names to run, should match keys of checkers dict (empty
list means run all)
51:     @param verbose     Verbosity of the output (0, 1, 2)
52:     @param criteria     Determines failure (lenient, normal, strict)
```

```
53: @param output_filename Path to the file for output. Use '-' for stdout.
54: @param skip_checks    Names of checks to skip
55: @param include_checks Names of checks to include
56: @param output_format  Format of the output(s) (e.g., 'text', 'html', 'json').
57:
58: @returns              Tuple indicating if the tests failed (based on the criteria) and if errors occurred.
59: """
```

Issue Report 2:

Please read and understand the following issue report:

Issue Title:

Fail result for Flags and flag attributes check with no reasoning

Issue Description:

Running the CF checker on [this file]

(http://thredds.aodn.org.au/thredds/fileServer/IMOS/SRS/SST/ghrsst/L3S-1d/dn/2016/20160919092000-ABOM-L3S_GHRSSST-SSTfnd-AVHRR_D-1d_dn.nc) results in

Name	Priority:	Score:Reasoning
------	-----------	-----------------

§3.5 Flags and flag attributes	:3:	31/32 :
--------------------------------	-----	---------

...

An apparently failed result, with no reasoning message.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- Go to the repository
- Click on Issues
- Click on **'New issue'**
- Copy and paste the content in the title and description sections
- Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/cf.py]

Buggy Methods:

[check_flags]

Recommended Fix:

Summary: Ensure reasoning messages are provided for all failed checks in check_flags function

File: compliance_checker/cf/cf_1_6.py

Function: check_flags

Lines: 1212-1218

Fix: Add reasoning message for failed flag check in check_flags function

Suggested Fix:

```
1212:         allvr = Result(BaseCheck.MEDIUM, allv, self.section_titles["3.5"])
1213:         if not allvr.value:
1214:             allvr.msgs = [
1215:                 f"flag masks and flag values for '{name}' combined don't equal flag values",
1216:                 "Ensure that the flag_masks and flag_values are correctly defined and compatible."
1217:             ]
1218:
1219:         ret_val.append(allvr)
```

Issue Report 3:

Please read and understand the following issue report:

Issue Title:

JSON output issue in 3.0.3 with ERDDAP URLs

Issue Description:

I've been testing the new Compliance Checker release with ERDDAP (on Windows), and I see some different results in the output when using ERDDAP URLs with the JSON output format than OPeNDAP.

I made a test script that illustrates the issue:

<https://gist.github.com/mwengren/68f61b0e70868a57336b81f863d7dbc9>

Or, I think you can see the same issue (in my environment at least) by running these commands:

compliance-checker -t cf -f json http://coastwatch.pfeg.noaa.gov/erddap/griddap/osuSstAnom

compliance-checker -t cf -f json http://ona.coas.oregonstate.edu:8080/thredds/dodsC/NANOOS/OCOS

The issue is that when I try to hit the ERDDAP URL, it includes what looks like a debug line in stdout:

Using cached standard name table v29 from C:\\Users\\Micah.Wengren\\.local\\share\\compliance-checker\\cf-standard-name-table-test-29.xml

```
{  
  "cf": {  
    "scored_points": 168,  
    "low_count": 0,
```

When I try to parse this into JSON using subprocess.Popen (see the Gist), it fails because it can't parse the first line.

I'm using 3.0.3:

\$ compliance-checker --version

IOOS compliance checker version 3.0.3

Also, not sure if it matters, but I'm using Python 2.7 for these tests.

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

[compliance_checker/cf/util.py, compliance_checker/cf/cf.py]

Buggy Methods:

[download_cf_standard_name_table]

Recommended Fix:

Summary: Fix JSON output issue by redirecting debug information to stderr to ensure JSON parsing is not disrupted.

File: cchecker.py

Function: main

Lines: 298-302

Fix: Redirect debug output to stderr when using JSON format

Suggested Fix:

```
298: if args.format != "json":
299:     print(
300:         f"Running Compliance Checker on the datasets from: {args.dataset_location}",
301:         file=sys.stderr,
302:     )
```

Issue Report 4:

Please read and understand the following issue report:

Issue Title:

JSON output missing argument

Issue Description:

Attempting to run compliance-checker on Python 2 or 3 with JSON output fails due to a missing function argument.

```
compliance-checker --test ioos -f json -c strict compliance_checker/tests/data/conv_bad.nc
```

Traceback (most recent call last):

```
File "/home/badams_local/.virtualenvs/cchecker_py3/bin/compliance-checker", line 9, in <module>
    load_entry_point('compliance-checker', 'console_scripts', 'compliance-checker')()
File "/home/badams_local/devel/compliance-checker/cchecker.py", line 41, in main
    args.format)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 50, in
run_checker
    groups = cls.json_output(cs, score_groups, output_filename, ds_loc, limit)
File "/home/badams_local/devel/compliance-checker/compliance_checker/runner.py", line 116, in
json_output
    cs.json_output(checker, groups, f, ds_loc)
TypeError: json_output() missing 1 required positional argument: 'limit'
```

After reading and understanding the issue, please complete the following instructions.

1. Create the issue on [INSIGHT_user_study_repository](#) by performing the following steps:

- (a) Go to the repository
- (b) Click on Issues
- (c) Click on **'New issue'**
- (d) Copy and paste the content in the title and description sections
- (e) Click **'Submit new issue'**

2. Observe INSIGHTS's suggestions for the bug localization and program repairing/bug resolving suggestions features. Feel free to take notes.

3. Assess the suggestions provided by INSIGHT. For your reference, we have included information about the actual buggy files and methods, as well as common approaches to fixing these bugs. When answering, please compare INSIGHT's suggestions with this reference information.

Buggy Files:

['compliance_checker/tests/test_cli.py', 'compliance_checker/suite.py']

Buggy Methods:

['TestCLI', 'CheckSuite']

Recommended Fix:

Summary: Fix TypeError in run_checker by adding missing 'output_type' argument to json_output call

File: compliance_checker/runner.py

Function: run_checker

Lines: 121-125

Fix: Fix missing argument in json_output call in run_checker function

Suggested Fix:

```
121:         elif out_fmt in {"json", "json_new"}:
122:             # Update file name if needed
123:             if len(output_format) > 1 and output_filename != "-":
124:                 output_filename = f"{os.path.splitext(output_filename)[0]}.json"
125:             cls.json_output(cs, score_dict, output_filename, ds_loc, limit, out_fmt)
```

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed file-level bug * localization for the created issues?

☒ Yes

☐ No

☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's file-level bug localization? *

- ☐ Very easy
- ☒ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

After inspecting INSIGHT's suggestions, do you think INSIGHT correctly performed method-level *
bug localization for the created issues?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

How easy is it to understand INSIGHT's method-level bug localization? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Please read the '**Technical Analysis**' section provided by INSIGHT, explaining the localization of the buggy files and methods. Do you think INSIGHT properly analyzed the context of the repository codebase while localizing the buggy files and methods?

- ☒ Yes
- ☐ No
- ☐ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis of the bug localization process? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

How easy is it to understand INSIGHT's suggested resolution of the identified bugs? *

- ☒ Very good
- ☐ Moderately good
- ☐ Neutral
- ☐ Moderately poor
- ☐ Very poor

Please read the '**Recommended Fix**' section suggested by INSIGHT. Do you think INSIGHT provided accurate suggestions to address/resolve the bugs? *

- ☐ Yes
- ☐ No
- ☒ Unsure

Please explain your answer (why yes, no, or unsure?)

.....

How easy is it to understand INSIGHT's provided analysis regarding the common ways to address the identified bugs? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately difficult
- ☐ Very difficult

Evaluating the Overall Experience with INSIGHT

How easy or difficult to use INSIGHT is? *

- ☒ Very easy
- ☐ Moderately easy
- ☐ Neutral
- ☐ Moderately hard
- ☐ Very hard

Please provide any suggestions for improving INSIGHT's Graphical User Interface (If any)

.....

Overall, how accurate the suggestions of INSIGHT's bug localization features and bug resolution *
features were?

- ☐ Very accurate
- ☒ Moderately accurate
- ☐ Neutral
- ☐ Moderately inaccurate
- ☐ Very inaccurate

Please explain your answer (optional).

.....

How helpful is INSIGHT in terms of localizing buggy code files and methods? *

- ☒ Very helpful
- ☐ Moderately helpful
- ☐ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How helpful is INSIGHT in terms of correctly suggesting strategies to fix the bugs? *

- ☐ Very helpful
- ☐ Moderately helpful
- ☒ Neutral
- ☐ Moderately unhelpful
- ☐ Very unhelpful

Please specify the reason for your answer (optional).

.....

How responsive is INSIGHT in terms of bug localization and providing suggestions to fix the bugs? *

- ☒ Very responsive
- ☐ Moderately responsive
- ☐ Neutral
- ☐ Moderately unresponsive
- ☐ Very unresponsive

What additional functionality/feature (if any) would you like to see in INSIGHT in the future? *

- ☐ Automatic patch suggestions
- ☒ Automatic PR generation with fixed patch
- ☒ Confidence scores and traceability (link to commits/highlight lines/methods)
- ☐ CI/CD integration (post findings on PRs, block risky merges, regression checks)
- ☐ IDE plugin implementation for in-context guidance and quick fixes while developing
- ☐ Triage assistance (severity/impact prediction, assignee suggestions)
- ☐ Other:

Please provide any (other) recommendations to improve INSIGHT

.....

Participant's Background and Experience

Please choose your highest credit obtained

- ☐ High School
- ☒ Bachelors
- ☐ Masters
- ☐ PhD
- ☐ Others

How many years of experience do you have in programming?

5

How many years of experience do you have in using Github?

4

How many years of experience do you have in managing and solving issue reports?

1

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